

A Stochastic Formulation to Telegrapher's Equation in Higher Dimensions

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I. INTRODUCTION

Efforts to govern lossy wave propagation with evolution of Random Walks (RW) dates back to Goldstein [1] and Kac [2]. Similar approaches were suggested and have recently been applied in related phenomena including diffusion process [3], transport process [4] and parabolic wave equation [5]. The motivation to represent governing Telegrapher's Equation (TE) with RW's is revealed in the nature of conventional full electromagnetic solutions required in applications of interest like scattering from obstacles and mode characterization of arbitrary shaped structures. The main simplification with RW's is that no meshing of the domain is required, thus the solution at points in space can be found independently from each other—allowing parallel computation.

Kac's telegrapher's process approach [2] gave insight by associating the loss in the medium to a Poisson process, the loss-less wave equation being treated by RW. In other words, given the exact solution to the wave equation, one can achieve the solution to TE by simply substituting the time dependence in the solution to the wave equation with a random quantity, which will be mentioned later. However, a priori solution of the wave equation is not available in complex geometries—the basis of random processes is to be constructed microscopically, as medium constituents moving with random direction changes. Nevertheless, Kac's way of treating the one-dimensional (1D) TE complies with the Persistent Random Walk (PRW) [6] model used for RW solutions to other phenomena and is explanatory for how to treat 1D wave propagation. Appropriate distributions were later defined as how to place partially reflecting or absorbing boundaries in 1D [7],[8]. However these efforts to satisfy TE fail in multi-dimensional media for reasons to become clear shortly. Similarly, the efforts stemming from PRW approach [4],[9] did not go beyond 1D as well. Appropriate PRW models to satisfy the cases of two-dimensional (2D) and higher dimensional TE have not been addressed up-to-date. The main focus of this work is to point out the deficiencies in PRW model in multi-dimensions and to backtrack and modify PRW constituents (probabilities of taking given directions).

We first complete the treatment of the 1D case by including the non-zero 'velocity' initial condition, a case not considered

in Kac's original work. This is discussed in section II. Section III is devoted to the PRW models necessary for multi-dimensional TE, 2D case in particular.

II. TELEGRAPHER'S PROCESS

The well-known TE which governs time-domain wave propagation in a lossy medium is:

$$\nabla^2 \psi(r, t) = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \psi(r, t) + \frac{2a}{c^2} \frac{\partial}{\partial t} \psi(r, t), \quad (1)$$

$\psi(r, t)$ the field value at spatial coordinates r and time t . Quantity a is associated with the loss in the medium and is inversely proportional to a relaxation time, where c is the speed of propagation. Consider the random time:

$$\tilde{t} = \int_0^t (-1)^{N(\tau)} d\tau, \quad (2)$$

where the randomness comes from $N(t)$ being a Poisson process with arrival rate a (1/seconds). For a Poisson process,

$$\text{Prob} \{N(t) = k\} = e^{-at} \frac{(at)^k}{k!}. \quad (3)$$

It is obvious that $-t < \tilde{t} < t$ in general and for $N(\tau) = 0$ in $(0, t)$, $\tilde{t} = t$. Kac considered (1) in 1D, where ∇ operator reduces to $\partial/\partial x$ along spatial variable x , as an Initial Value Problem (IVP) with non-zero initial distribution $\psi(x, t = 0) =: \phi(x)$ and zero initial derivative distribution ('velocity' distribution) $\frac{\partial}{\partial t} \psi(x, t)|_{t=0} = 0$. It was shown that the exact solution to (1) in 1D with these initial conditions is recovered by the averaging of an ensemble of random solutions [2]:

$$\psi(x, t) = \frac{1}{2} \langle \phi(x - c\tilde{t}) + \phi(x + c\tilde{t}) \rangle, \quad (4)$$

where $\langle . \rangle$ is the averaging operator. Note that the solution to the wave equation is $\psi(x, t) = \frac{1}{2} [\phi(x - ct) + \phi(x + ct)]$. Now consider the case with initial conditions $\psi(x, t = 0) = 0$ and $\frac{\partial}{\partial t} \psi(x, t)|_{t=0} =: \beta(x)$. Since (1) actually consists of two coupled equations, just like voltage and current equations in the case of a lossy transmission line, it can be shown that the solution to the latter IVP is:

$$\psi(x, t) = \frac{1}{2} \langle \zeta(x + c\tilde{t}) - \zeta(x - c\tilde{t}) \rangle, \quad (5)$$

where $\zeta(x) = \frac{1}{c} \int_{-\infty}^x \beta(\lambda) d\lambda$. Assuming the initial distribution ceases at $x = \pm\infty$, we have:

$$\psi(x, t) = \frac{1}{2c} \left\langle \int_{x-c\tilde{t}}^{x+c\tilde{t}} \beta(\lambda) d\lambda \right\rangle. \quad (6)$$

Note that (4) and (6) are respective stochastic solutions to (1) in 1D as the two IVP's considered separately. Superposing the two IVP's, i.e. adding (4) and (6), one gets an expression similar to the well known solution to the 1D TE with initial conditions $\phi(x)$ and $\beta(x)$. This is as if the time dependency in the d'Alembert's solution [10] of a corresponding wave equation is replaced with the random quantity $\tilde{t}$ in (2) and the ensemble of random distributions are averaged. The number of realizations in the ensemble should be large enough for the solutions to converge. The requirement $at \gg 1$ should also be satisfied so that the Poisson process has fairly enough number of arrivals in $[0, t)$.

A test case for the IVP with $\phi(x) = 0$ and non-zero $\beta(x)$ is considered next. The analytical solution to the 1D TE in (1) and the stochastic solution in (6) give the same final distribution as depicted in Figure 2. The sinusoidal $\beta(x)$ depicted in Figure 1 spreads along space as time progresses in $[0, T]$, hence some of the initial energy does not reside in the region $[-cT, cT]$ in Figure 2. This is simply the result of getting non-zero $\tilde{t}$ in a realization which spreads the stochastic solution on both sides, known as the waking effect in the exact analytical solution of TE [11].

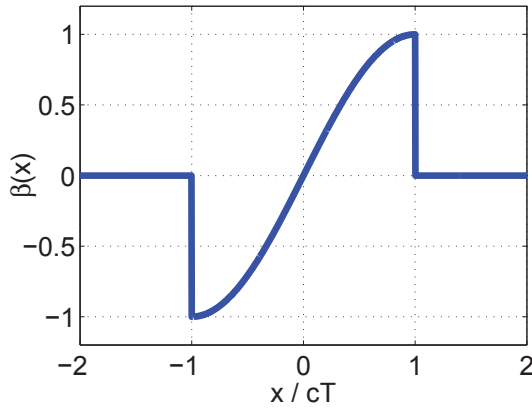


Fig. 1. Initial distribution for the derivative function. The parameters $c=1$ and $T=1$.

III. PRW APPROACH FOR 2D

Like the ordinary RW, the PRW is constructed microscopically starting from probabilities of traversing a given direction

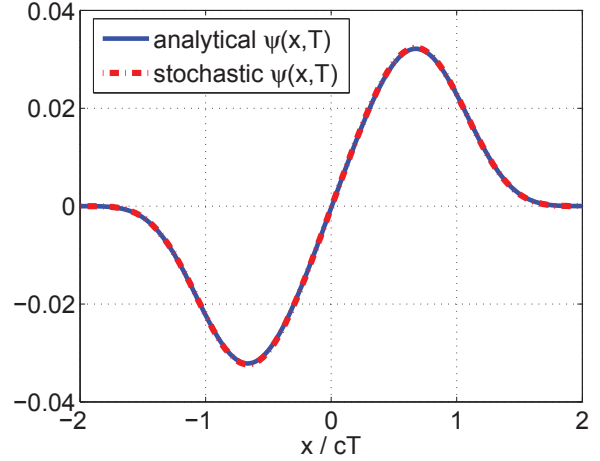


Fig. 2. Analytic and stochastic solutions at time T for the initial distribution depicted in Figure 1. The parameters $c=1$, $a=10$, $T=1$ and number of realizations $N_r=100$.

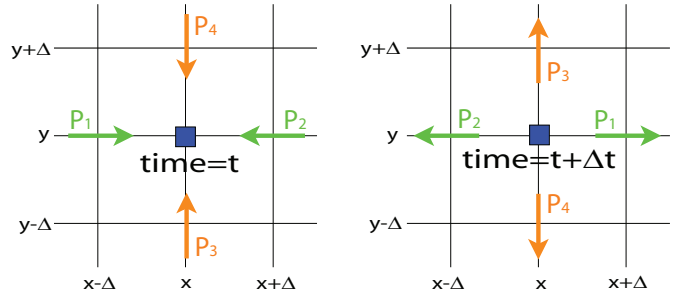


Fig. 3. The PRW model for probabilities of taking designated directions, e.g. P_1 is the probability a particle is in $+x$ direction and P_4 in $-y$ direction. The diagram on left is at time t , the one on right is at time $t+1$.

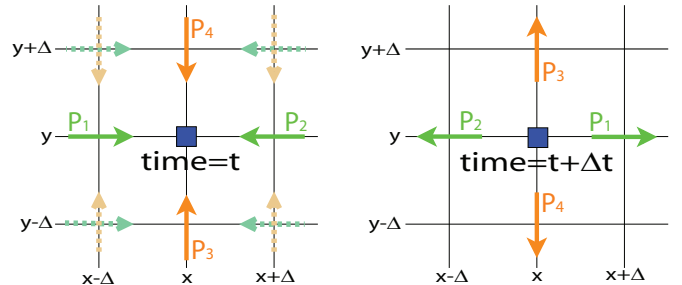


Fig. 4. The modified PRW model. Probabilities P_1 through P_4 are not solely system constituents anymore but are excitations. The dashed probabilities are utilized while completing the model to satisfy 2D TE.

along a coordinate axis and taking the continuum limit to govern the a partial differential equation, TE in our case. The validity of 1D PRW model for governing TE in the continuum limit was shown [12], [4] and appropriate boundary conditions were defined [13]. Unlike 1D however, non-validity of this model is also known [12], [4]. Very recently, Plyukhin [14] suggested a recipe to handle the non-validity of the model in the continuum limit in dimensions higher than one. We will backtrack the modified model and find out the necessary microscopic view of this recipe in this section.

Consider the 2D model depicted in Figure 3. P_1 through P_4 denote the probabilities of a particle moving in the denoted direction at the given discrete time step. The speed of particles is constant $c = 2\Delta/\Delta t$, where an event always occurs at the half time of the current step. At each time step, one of three events occur: the particle keeps moving in its previous direction with probability T , reverses its direction with probability R or scatters to either of the two orthogonal directions with probability L each. Therefore, the total probability at each time step satisfies $T + R + 2L = 1$. The relation between probabilities at the two time steps depicted in Figure 3 is therefore:

$$\begin{bmatrix} P_1^{1,0}(t + \Delta t) \\ P_2^{-1,0}(t + \Delta t) \\ P_3^{0,1}(t + \Delta t) \\ P_4^{0,-1}(t + \Delta t) \end{bmatrix} = \begin{bmatrix} T & R & L & L \\ R & T & L & L \\ L & L & T & R \\ L & L & R & T \end{bmatrix} \times \begin{bmatrix} P_1^{-1,0}(t) \\ P_2^{1,0}(t) \\ P_3^{0,-1}(t) \\ P_4^{0,1}(t) \end{bmatrix}, \quad (7)$$

where $P_k^{i,j}(t) = P_k(x + i\Delta, y + j\Delta, t)$, $k = 1, \dots, 4$. Now, as each of these linear equations is expanded in first order Taylor approximation, the following equation is obtained in the continuum limit:

$$\left[\frac{\partial^2}{\partial t^2} + \frac{2(R+L)}{\Delta t} \frac{\partial}{\partial t} \right] (\rho_x + \rho_y) = c^2(1 - R - L) \times \left(\frac{\partial^2 \rho_x}{\partial x^2} + \frac{\partial^2 \rho_y}{\partial y^2} \right) \quad (8)$$

where $\rho_x = P_1 + P_2$ and $\rho_y = P_3 + P_4$ are the probabilities of a particle traversing in x-axis and y-axis, respectively. The above equation is not what we sought to arrive at-it is not 2D TE. First of all, the limits $(R,L) \rightarrow (0,0)$, $\Delta t \rightarrow 0$, $(R+L)/\Delta t \rightarrow \theta > 0$, were imposed as PRW limits. In that case it is similar to a TE in 2D except for the missing cross derivative terms in the last term. If the last term in the parenthesis contained the missing terms to complete the Laplacian operator in 2D, then this equation would satisfy TE in $(\rho_x + \rho_y)$. This was pointed out in [14]. This is a modification in the physics in microscopic sense; P_1, P_2, P_3 and P_4 will not be simple constituents, but rather will be excited as time progresses in the modified case. We will now backtrack the 2D TE without changing limits and

form a new linear system of the four states that would have resulted in 2D TE.

As (8) is completed to 2D TE in $(\rho_x + \rho_y)$, it is easy to see that with the same PRW limits the desired 2D TE stems from the following modified set of relations which is analogous to (7):

$$\begin{bmatrix} P_1^{1,0}(t + \Delta t) \\ P_2^{-1,0}(t + \Delta t) \\ P_3^{0,1}(t + \Delta t) \\ P_4^{0,-1}(t + \Delta t) \end{bmatrix} = \begin{bmatrix} T & R & L & L \\ R & T & L & L \\ L & L & T & R \\ L & L & R & T \end{bmatrix} \times \begin{bmatrix} P_1^{-1,0}(t) \\ P_2^{1,0}(t) \\ P_3^{0,-1}(t) \\ P_4^{0,1}(t) \end{bmatrix} + \frac{1}{2} \times \begin{bmatrix} P_3^{-1,-1}(t) - P_3^{1,-1}(t) + P_4^{-1,1}(t) - P_4^{1,1}(t) \\ -P_3^{-1,-1}(t) + P_3^{1,-1}(t) - P_4^{-1,1}(t) + P_4^{1,1}(t) \\ P_1^{-1,-1}(t) - P_1^{1,-1}(t) + P_2^{-1,1}(t) - P_2^{1,1}(t) \\ -P_1^{-1,-1}(t) + P_1^{1,-1}(t) - P_2^{-1,1}(t) + P_2^{1,1}(t) \end{bmatrix}. \quad (9)$$

The last term basically modifies the microscopic relations dependent on the medium excitations. In other words, the probability of the particle moving in positive x direction for instance at time $t + \Delta t$ depends not only on the probabilities at time t , but the x derivatives of P_3 and P_4 at t as well. This relation between the states in (9) is depicted in Figure 4. The dashed arrows in Figure 4 denote the probabilities at time t that are completing the desired model to satisfy 2D TE and are those probabilities in the last term in (9).

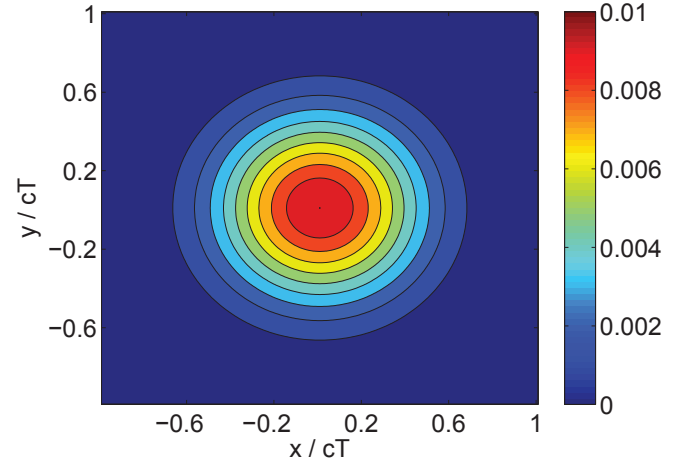


Fig. 5. The free-space Green's function solution for an impulsive source excited from the origin of the xy-plane at time $t = 0$. The parameters $a = 0.2 \text{ s}^{-1}$, $c = 2 \text{ m/s}$, observation time $T = 50 \text{ s}$.

IV. CONCLUSION

Accuracy of Kac's telegrapher's process approach for the non-zero velocity was demonstrated in II. Although this process can recover 1D TE and other boundary conditions for this approach can be developed, the approach could not be applied in higher dimensions. Although Kac's randomized

wave equation solution meets analytical TE solution in higher dimensions as well, we did not mention it since it does not avoid solving the wave equation which is no easier than solving TE.

The PRW on the other hand also governs 1D TE in the continuum limit, but not in 2D. We showed that Plyukhin's modification to the state transitions reads (9), i.e. the probabilities of states at a time step depend also on cross derivative terms at the previous step. A test case to validate the modified PRW model in (9) is the simple 2D wave propagation in free-space. The free-space Green's function solution for an impulse excited at time $t = 0$ from the origin of the xy-plane is demonstrated in Figure 5. Corresponding stochastic solution to the above case along with the case with partially reflecting/absorbing boundary conditions will be demonstrated at the symposium.

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